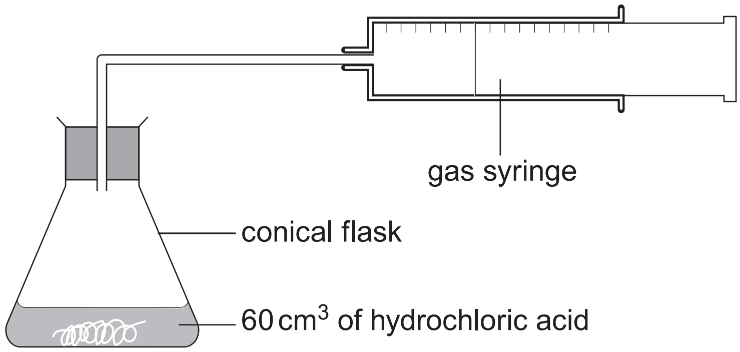
4. Magnesium ribbon was reacted with excess dilute hydrochloric acid. The volume of hydrogen produced during the reaction was measured over time using a gas syringe. The experiment was carried out at a temperature of 20 °C.



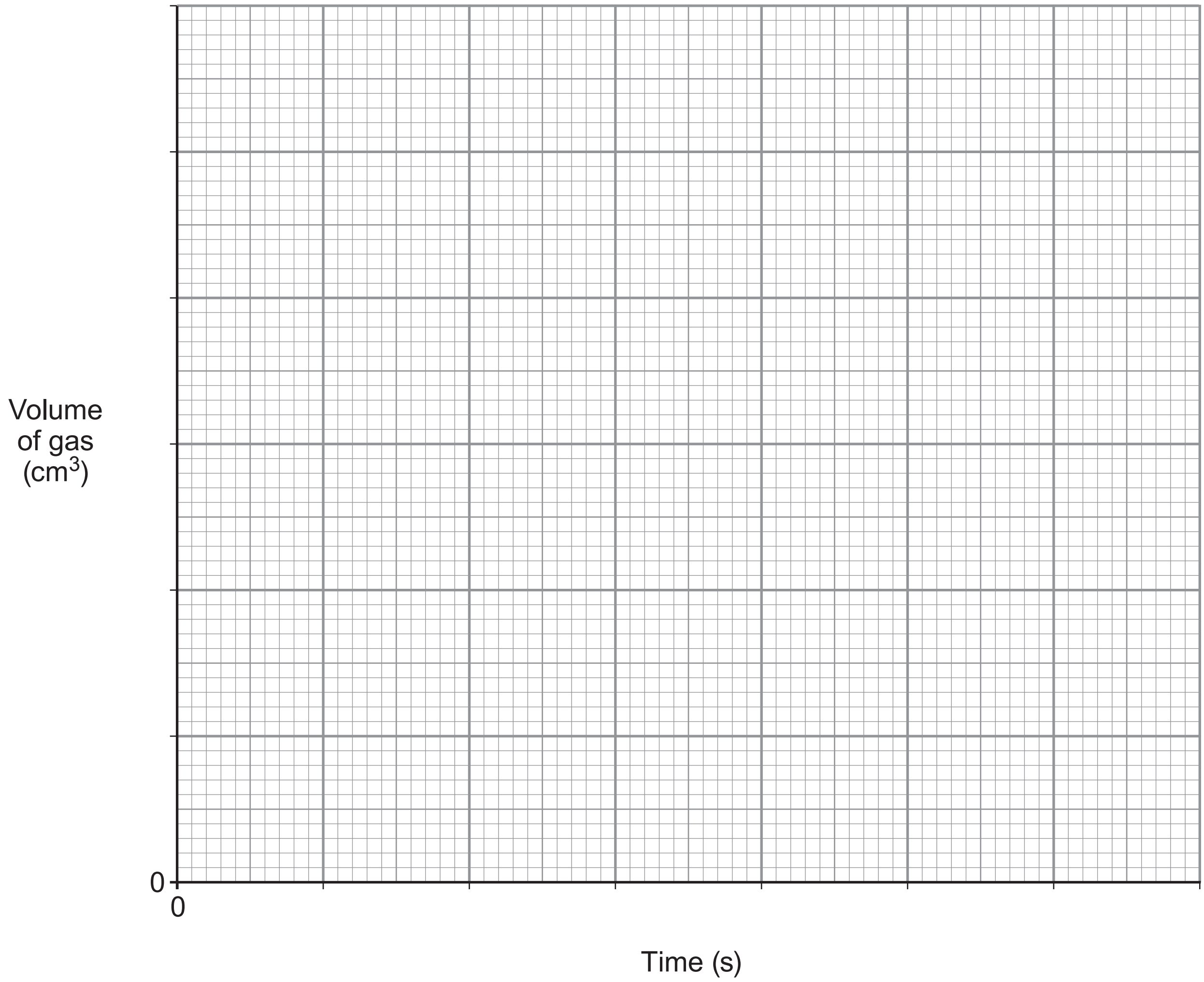
The table shows the results obtained in the experiment.

Time (s)	Volume of gas (cm^3)
0	0
10	34
20	64
30	78
40	94
50	98
60	100
70	100

(a) Plot the data **on the grid opposite** and draw a suitable line. Label the graph **A**. [3]

(b) **On the same grid**, sketch the graph that would be obtained if a piece of magnesium ribbon of half the length were reacted with another 60 cm^3 of the same hydrochloric acid. Label this graph **B**. [2]

Examiner only



(c) If the original experiment were repeated at a higher temperature, the reaction would occur at a higher rate. Explain this using particle theory. [2]

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(d) This method can be used to investigate the rate of this reaction. Without changing the method or apparatus used, state what should be done to improve the overall strength of the evidence collected. [2]

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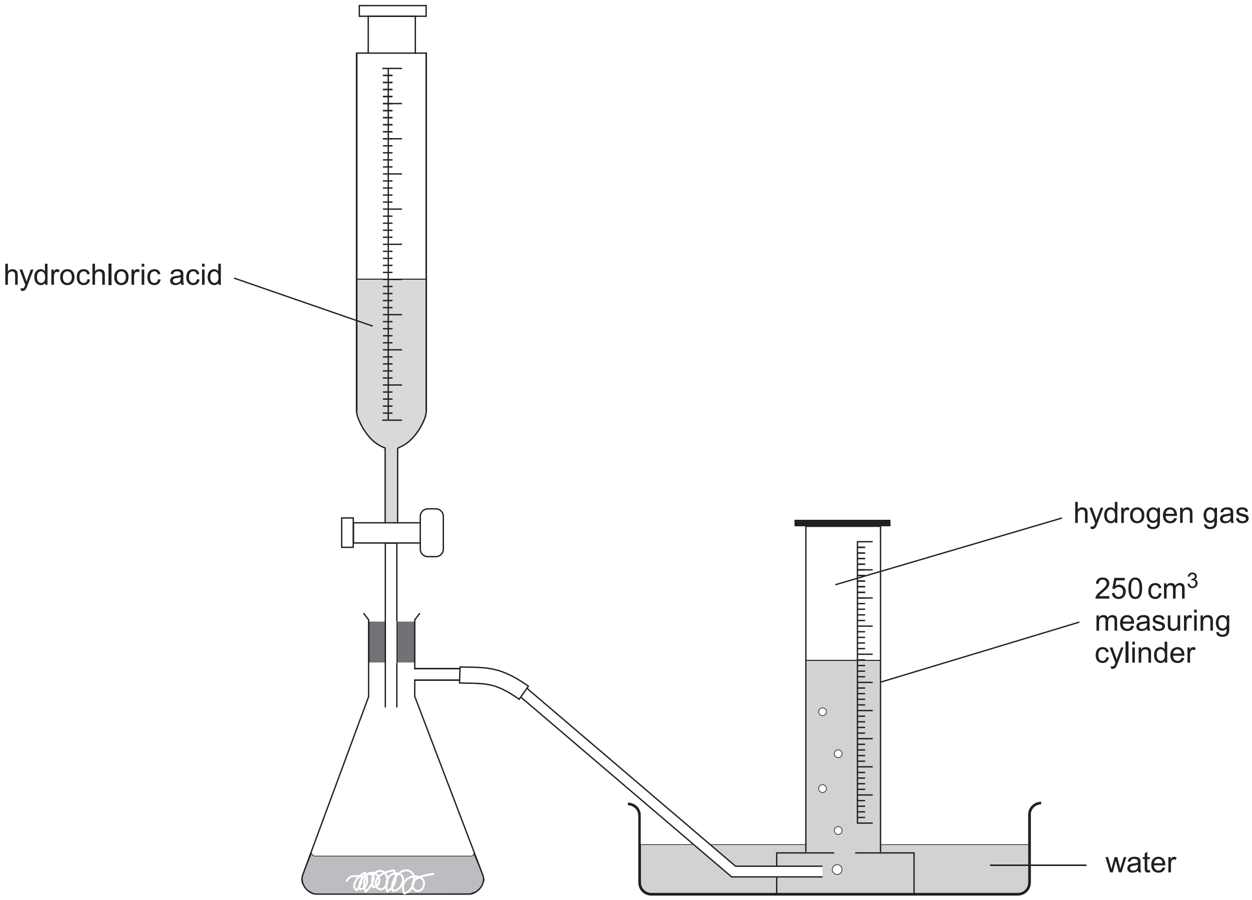
Turn over.

Examiner only

(e) Give an advantage and a disadvantage of using the following alternative apparatus to collect the hydrogen gas. [2]



Advantage ....

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Disadvantage ....

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